SUPPOSITIONS

STANDARD PROOF TECHNIQUES IN DISCRETE MATHEMATICS AND EXAMPLE APPLICATIONS

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ABSTRACT

This document examines standard proof techniques found in discrete mathematics. It combines meticulous explanations with example applications to accessibly define and demonstrate each approach.

CONTENTS

1 Proof Techniques for Existential Statements 1
1.1 Proof by Construction 2

2 Proof Techniques for Universal Statements 3
2.1 Proof by Exhaustion 4
2.2 Direct Proof 5
2.3 Proof by Division into Cases 6
2.4 Proof by Contraposition 7
2.5 Proof by Contradiction 8
2.6 Proof by Mathematical Induction 9
2.7 Proof of Subset Containment 10
2.8 Proof of Injectivity 11
2.9 Proof of Surjectivity 12
2.10 Proof of Bijectivity 13

3 Example Problems 14
3.1 Squares of Odd Integers 15
3.2 Distances from a Fixed Point 16
3.3 Lattice Points in Circles 17
3.4 Dot Products of Factorials 18
3.5 Left Inverses of Injections 19
3.6 Intersections of Open Sets 20

1 PROOF TECHNIQUES FOR EXISTENTIAL STATEMENTS

1.1 PROOF BY CONSTRUCTION

2 PROOF TECHNIQUES FOR UNIVERSAL STATEMENTS

2.1 PROOF BY EXHAUSTION

2.2 DIRECT PROOF

2.3 PROOF BY DIVISION INTO CASES

2.4 PROOF BY CONTRAPOSITION

2.5 PROOF BY CONTRADICTION

2.6 PROOF BY MATHEMATICAL INDUCTION

2.7 PROOF OF SUBSET CONTAINMENT

2.8 PROOF OF INJECTIVITY

2.9 PROOF OF SURJECTIVITY

2.10 PROOF OF BIJECTIVITY

3 EXAMPLE PROBLEMS

3.1 SQUARES OF ODD INTEGERS

Exercise. Prove that if a is an odd integer, then $a^2 = 8m + 1$ for some $m \in \mathbb{Z}$.

Proof. Hello! □

3.2 DISTANCES FROM A FIXED POINT

Exercise. Prove that if $x < -4$ and $y > 2$, then the distance from (x, y) to $(1, -2)$ is at least 6.

Proof. Hello! □

3.3 LATTICE POINTS IN CIRCLES

Exercise. A point $(m, n) \in \mathbb{R}^2$ is a lattice point if $m, n \in \mathbb{Z}$. Prove that the number of lattice points inside any circle centered at the origin equals $4k + 1$ for some $k \in \mathbb{Z}$.

Proof. Hello! □

3.4 DOT PRODUCTS OF FACTORIALS

Exercise. For $(a_1, a_2, \ldots, a_n), (b_1, b_2, \ldots, b_n) \in \mathbb{R}^n$, the dot product is

$$(a_1, a_2, \ldots, a_n) \cdot (b_1, b_2, \ldots, b_n) = a_1 b_1 + a_2 b_2 + \cdots + a_n b_n.$$

For example, $(5, 2, -3) \cdot (3, 7, 4) = 5 \cdot 3 + 2 \cdot 7 + (-3) \cdot 4 = 17$. Prove that

$$(1, 2, 3, \ldots, n) \cdot (1!, 2!, 3!, \ldots, n!) = (n+1)! - 1$$

for all positive integers n.

Proof. Hello! □

3.5 LEFT INVERSES OF INJECTIONS

Exercise. Let A be a nonempty set. Prove that a function $f\colon A \to B$ is injective if and only if f has a left inverse, that is, a function $g\colon B \to A$ such that $(g \circ f)(x) = x$ for all $x \in A$.

Proof. Hello! □

3.6 INTERSECTIONS OF OPEN SETS

Exercise. A set $S \subseteq \mathbb{R}$ is open if for each $b \in S$, there exists $\epsilon > 0$ such that $(b-\epsilon, b+\epsilon) \subseteq S$. Prove that the intersection of two open sets is open.

Proof. Hello! □